

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-071

WASTE DISCHARGE REQUIREMENTS
FOR
CITY OF MARYSVILLE
WASTEWATER TREATMENT AND RECLAMATION FACILITY
YUBA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. The City of Marysville (hereafter Discharger) previously submitted a Report of Waste Discharge, dated 28 January 1993, an Initial Study/Negative Declaration, dated October 1991, and a document titled *City of Marysville, Final Report, Wastewater Facilities Plan* dated January 1992, for their wastewater treatment, reclamation and disposal facility.
2. The Marysville Wastewater Treatment Plant (Plant), Assessor's Parcel Number (APN) 010-260-019, disposal ponds APN's 010-260-017 and 020-020-002 and reclamation areas (hereafter Designated Reclamation Areas) APN's 010-013-001 and -002, 010-014-004, 010-012-001 and -002 are all on property owned by the Discharger. The walnut orchard, APN 20-010-10, adjacent to the disposal ponds east of the Yuba River is also a Designated Reclamation Area, and is owned by Bruce and Mary Jenkins. The walnut orchard is regulated under a separate permit.
3. Waste Discharge Requirements (WDRs) Order No. 95-093, adopted by the Board on 28 April 1995, prescribes requirements for a discharge from a secondary treatment plant to evaporation/percolation disposal ponds (ponds) and walnut orchard irrigation and tertiary treated effluent to existing soccer fields. The old on-site ponds (50 acres installed 1962) are situated on the land immediately north of the Yuba River and Feather River confluence.
4. Twenty-two acres of additional ponds were installed in 1989 across the Yuba River. A 35 acre walnut orchard is located adjacent to the new ponds southeast of the facility across the Yuba River. There are a total of 33 evaporation/percolation ponds at both sites. On-site and off-site ponds are protected by levees from a 10-year flood event.
5. A pipeline was constructed underneath the Yuba River to transport wastewater to the off-site ponds. Two flow meters were installed; one measures flow on the plant side and the second measures flow across the river on the pond side. Any significant difference between measured flow from these two meters would be an indication of possible pipeline leakage. Presently, there is no on-site ponds' flow meter and the plant side off-site ponds' flow meter is inoperable.
6. Order No. 95-093 is neither adequate nor consistent with current plans and policies of the Board. These WDRs are being updated and revised to reflect current plans and policies of the Board.

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7. The Marysville treatment, disposal and reclamation facilities are in Sections 23 and 24, T15N, R3E, MDB&M, with surface water drainage to the Feather and Yuba Rivers, as shown in Attachment A, which is attached hereto and part of this Order by reference. The on-site and off-site evaporation/percolation ponds, soccer fields, and walnut orchard are designated disposal areas.
8. The facility lies within the Marysville hydrologic unit/area/subarea (No. 515.20 and 515.40), as depicted on the interagency hydrologic maps prepared by the Department of Water Resources in August 1986.
9. The existing treatment plant consists of one bar screen, two grit channels with chain and flight removal, two primary clarifiers, two trickling filters, one secondary clarifier, tertiary clarifier, filter, chlorinator, and holding pond, two anaerobic digesters, and sludge drying beds. Currently, disposal of wastewater is accomplished via discharge to evaporation/percolation ponds and the walnut orchard.
10. The Discharger has constructed a new tertiary treatment facility which will produce an effluent meeting Title 22 reclamation criteria standards for discharge to existing and proposed soccer fields. The new tertiary treatment unit (TTU) consists of two clarifier/filter units, chlorination, and holding ponds. An existing pond is proposed to be used as a reclaimed water holding pond prior to irrigation, and would overflow into the existing ponds when irrigation demands are low. The TTU system has a design capacity of 0.8 million gallons per day (MGD).
11. The components of the treatment plant have various capacities. The components limiting the treatment capacity appears to be the disposal ponds, which have an estimated capacity of 1.7 MGD. Discharges to ponds averaged 1.5 MGD during the dry months of 1994 and 1.6 MGD March 1999 to May 2000. The Discharger reports current population of 12,854 and 3,633 sewer hookups within the service area.
12. The TTU will reduce the demand on the disposal ponds during periods of dry weather, which will in turn increase their wet weather effluent storage capacity. These waste discharge requirements prohibit the monthly average dry weather discharge flow to disposal ponds, walnut orchard, and soccer fields from exceeding 1.7 MGD.
13. For the purposes of this Order, "spray irrigation" means application of reclaimed water to crops by sprinklers and "surface irrigation" means application by flood, furrow, or border check irrigation.
14. The Discharger is the Producer of the reclaimed wastewater and Bruce and Mary Jenkins are the Users of the reclaimed wastewater for reuse on the walnut orchard. The Water Recycling Requirements (WRRs) for Bruce and Mary Jenkins is described in WRR Order No. 5-01-052.
15. The disposal of reclaimed wastewater at the walnut orchard has been historically accomplished with flood irrigation. Applications occur approximately 2 to 3 times each year during the months of July and August. Each application normally takes 2 to 3 days. Irrigation berms are

constructed to ensure that no wastewater is allowed to runoff during application. Berms are removed and 45 days must pass after the last wastewater application to allow walnut harvesting. The walnuts are picked up after being shaken to the ground from their trees.

16. The Department of Health Services (DHS) issued a memorandum to the Board dated 30 October 2000 recommending effluent standards for walnut orchards. The DHS has determined that disinfected, oxidized (secondary) effluent would be suitable for use for surface irrigation of an orchard where walnuts would be shaken to the ground for subsequent harvesting, if all of the following are obtained:
 - a. Reclaimed water will not be used within 45 days before walnuts are shaken to the ground;
 - b. The grower will inform the recipient that the walnuts may possibly contain sewage-borne organisms on the hulls;
 - c. The walnuts will be handled in a good hygienic manner that will prevent contamination of shells when the hulls are removed, and that will not cause dust or dirt to enter the shell to contaminate the walnut meat; and
 - d. The shell will be cleaned by a procedure that will meet the California Food Sanitation Act (Health and Safety Code, Division 22, Chapter 7) before they are offered for sale as human food.

There is no exception from the quality requirement of Section 60603 of Title 22 CCR, for reclaimed water which will be used for spray irrigation in an orchard.

17. The Discharger monitors seven groundwater monitoring wells (MWs): four MWs define groundwater in the vicinity of the on-site ponds east of the Feather River and west of the Yuba River immediately above the confluence, three MWs surround the off-site ponds situated across the Yuba River on its east bank. The monitoring well locations are shown on Attachment A, which is attached hereto and part of this Order by reference.
18. The Board has considered anti degradation pursuant to State Water Resources Control Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, Provisions contained within this Order and a Monitoring and Reporting Program included with this Order incorporates groundwater monitoring for data collection to determine whether the discharge will cause an increase in groundwater constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to implement source control, change the method of disposal, cease the discharge or take other action to prevent groundwater degradation.

19. The Board adopted a Water Quality Control Plan, Fourth Edition, for the Sacramento River and the San Joaquin River Basins (hereafter Basin Plan), which designates beneficial uses and water quality objectives for waters of the Basins. These requirements implement the Basin Plan.
20. The beneficial uses of the Feather and Yuba Rivers are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; and preservation and enhancement of fish, wildlife and other aquatic resources.
21. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
22. In October 1991 the City of Marysville certified an Initial Study/Negative Declaration in accordance with the California Environmental Quality Act (CEQA), (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines.
23. The action to revise and update waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA), in accordance with Title 14, California Code of Regulations (CCR), Section 15301.
24. This discharge is exempt from the requirements of *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste*, as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq., (hereafter Title 27). The exemption, pursuant to Section 20090(b), is based on the following:
 - a. The Board is issuing waste discharge requirements;
 - b. The discharge complies with the Basin Plan; and
 - c. The waste material does not need to be managed according to Chapter 11, Division 4.5, Title 22, of the CCR as a designated or hazardous waste.
25. Pursuant to Section 13521 of the California Water Code, the Department of Health Services has established statewide recycling criteria in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22) for the use of reclaimed water and has developed guidelines for specific uses.
26. The Board consulted with the Department of Health Services, County Health Department, and the Feather River Vector Control District and considered their recommendations regarding public health aspects for use of reclaimed water.
27. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
28. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 95-093 is rescinded and the City of Marysville, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereby, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. By-pass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous,' as defined in Sections 2520 & 2521(a) of Chapter 15, or 'designated,' as defined in Section 13173 of the California Water Code, is prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a pollution or nuisance as defined by the Porter-Cologne Water Quality Control Act, Section 13050.
2. The monthly average dry weather discharge flow shall not exceed 1.7 million gallons per day (MGD).
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and reclamation disposal areas.
4. As a means of discerning compliance with Discharge Specification No. 3, the dissolved oxygen content in the upper zone (1 foot) of the wastewater ponds and storage ponds shall not be less than 1.0 mg/l.
5. The discharge shall not cause degradation of any water supply.
6. The wastewater shall remain in the designated treatment and disposal areas at all times.
7. Public contact with wastewater shall be precluded through such means as fences, placards, and irrigation management practices.
8. Signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal to alert the public of the use of treated wastewater.
9. The facility and designated disposal areas except for the walnut orchard and soccer fields shall be designed, constructed, operated, and maintained to prevent inundation or washout due to flood and/or storm with a 100-year return frequency. Provision F.8 requires the Discharger to submit a technical report which provides an alternatives evaluation, cost, implementation plan, and time schedule for protection of the facility and

designated disposal areas from inundation or washout from flood and storm with a 100-year return frequency.

10. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the non-irrigation season and collection of all tailwater from the walnut orchard, if any, to comply with Discharge Specification B.11. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100-years, distributed monthly in accordance with historical rainfall patterns.
11. Pond freeboard shall never be less than two feet (measured vertically to the lowest point of overflow).
12. Before 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specifications Nos. 10 and 11.
13. The effluent from the treatment facility to the evaporation/ percolation ponds shall not exceed the following limits:

<u>Constituents</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40	80
Settleable Solids	ml/l	0.2	0.5
Total Nitrogen	mg/l as N	40	
pH	pH units	6.5 to 8.5	

¹ 5-day, 20° Celsius Biochemical Oxygen Demand

14. Ponds shall be managed to prevent breeding of mosquitoes. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

C. Reclaimed Wastewater:

1. The Discharger shall comply with applicable regulation criteria set forth in Title 22, Division 4, Section 60301 et seq., CCR.
2. The discharge of reclaimed wastewater to the Designated Reclamation Area (soccer fields) shall not exceed the following limitations:

<u>Constituents</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Average</u>	<u>Weekly Median</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	10 ²			30 ²
Total Suspended Solids	mg/l	10			30
Total Coliform Organisms	MPN/100 ml			2.2	23
Turbidity ³	NTU		2		5 ⁴

¹ 5-day, 20°C biochemical oxygen demand

² To be determined by a 24-hour composite

³ To be met prior to disinfection

⁴ Not more than 5% of the time during any 24-hour period

3. The discharge of reclaimed wastewater to the Designated Reclamation Area (walnut orchard) shall not exceed the following limitations:

<u>Constituents</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Average</u>	<u>Weekly Median</u>	<u>Daily Maximum</u>
BOD ¹	mg/l	40 ²			80 ²
Total Suspended Solids ³	mg/l	40			80
Total Coliform Organisms	MPN/100 ml			2.2	23

¹ 5-day, 20°C biochemical oxygen demand

² To be determined by a 24-hour composite

³ To be met prior to disinfection

4. Reclaimed water shall be applied to the soccer fields such that there shall be no standing water in the soccer fields 24 hours after wastewater application has ceased. Reclaimed water application to soccer fields shall be managed to minimize erosion and runoff.
5. Areas irrigated with reclaimed water shall be managed to prevent breeding of mosquitoes. Specifically:
- Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.
 - Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.

6. Direct or windblown spray shall be confined to the designated disposal areas and prevented from contacting drinking water facilities.
7. The discharger may not spray irrigate effluent during periods of precipitation and for at least 24 hours after cessation of precipitation when ground is saturated or when winds exceed 30 mph.
8. Application of reclaimed water to the designated reclamation disposal areas shall be at reasonable rates considering the crop, soil, climate, and irrigation management system. The nutrient loading of the reclamation area, including the nutritive value of organic chemical fertilizers and of the reclaimed water shall not exceed the crop demand.
9. There shall be no impoundment of wastewater or reclaimed water within 500 feet of any domestic water well, 100 feet of any irrigation well, 100 feet of any residential area, or within 50 feet of a watercourse unless it is demonstrated protective of public health and water quality to the satisfaction of Department of Health Services and the Executive Officer.
10. Designated reclamation disposal areas wastewater land disposal shall be limited to walnut orchard irrigation and soccer fields and shall comply with the Department of Health Services' *UNIFORM GUIDELINES FOR WASTEWATER DISINFECTION*. All mixed wastewater and storm water from the irrigation field must be returned to the storage pond throughout the entire year.
11. The Discharger shall conduct periodic inspections of the Users to monitor compliance with the criteria established in Title 22 and the requirements of this Order.
12. Reclaimed water controllers, valves, etc., shall be affixed with reclaimed water warning signs, and the quick couplers and sprinkler heads shall be of a type, or secured in a manner, that permits operation by authorized personnel only.
13. Supplementing reclaimed water by connection with irrigation, industrial, or domestic drinking water sources requires air gap devices.
14. Score cards used for all sports activities on the soccer field shall indicate that reclaimed water is used.
15. If the Discharger intends to use or produce reclaimed wastewater for crops other than the walnut orchard and for use other than the soccer fields, it shall first submit a written report demonstrating, to the satisfaction of the Executive Officer, that management of reclaimed water and irrigated properties will assure compliance with the terms of this Order.

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YUBA COUNTY

D. Sludge Disposal:

1. The Discharger shall dispose of all collected screenings, sludge, grease, oil, and any other solids removed from liquid wastes off site or in a manner that is consistent with CCR, Title 27 and approved by the Executive Officer.
2. Disposal of sludge shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR part 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in the 40 CFR part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR part 503 whether or not they have been incorporated into this Order.

3. Any proposed change in sludge use or disposal practice from a previously approved practice should be reported to the Executive Officer and US Environmental Protection Agency Regional Administrator at least 90 days in advance of the change.
4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*.

E. Groundwater Limitations:

1. The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number (MPN) of total coliform organisms to exceed 2.2 per 100 ml over any 7-day period.

F. Provisions:

1. By 1 May 2001, the Discharger shall comply with the State Water Resources Control Board Water Quality Order No. 97-03-DWQ (as amended), the *General Permit for Discharges of Storm Water Associated with Industrial Activities*, by either filing a Notice of Intent or a Notice of Non-Applicability.
2. By 1 June 2001 the Discharger shall provide a technical report on the field measurement accuracy and precision for plant influent flow, flow to the evaporation/percolation ponds, flow to the walnut orchard, flow to the tertiary irrigation system, and all pond freeboard measurements. A California registered professional civil engineer experienced in the field of wastewater treatment and disposal shall prepare this report.

3. By **2 July 2001**, the Discharger shall submit to the Board a technical report detailing a corrective action and contingency plan to be taken when treatment plant effluent quality fails to meet required standards or limitations.
4. By **2 July 2001**, the Discharger shall submit an updated sludge disposal plan describing the annual volume of sludge generated by the plant and specifying the disposal practices.
5. By **2 July 2001**, The Discharger shall maintain an Operations and Maintenance (O & M) Manual. Operating personnel shall keep a copy of the O & M Manual at the wastewater treatment plant site for reference. Operating personnel shall be familiar with its contents and location, and it shall be made available for review during Regional Board inspection. Items to be kept with the O & M Manual are as follows:
 - a. Operating procedures for plant wastewater treatment processes, irrigation, aeration, and chlorination (equipment runtimes, chlorine barrel replacement schedule) plus monitoring procedures, times, and locations.
 - b. Operation and maintenance records to include equipment maintenance schedules, operator logs, Monitoring and Reporting Program self-monitoring reports, system upsets, sludge disposal, and personnel training.
 - c. An emergency response plan to include reporting procedures, agency contact phone numbers, a spill response plan for hazardous materials, raw wastewater, partially treated wastewater, spill cleanup, and cleanup waste products disposal plan.
 - d. A copy of contingency plans defined in Provision F.1.
 - e. A copy of this Order.
6. By **3 September 2001**, the Discharger shall submit a report on the commercial and/or industrial dischargers within the service area. A discussion of possible Upset, Interference, Pass Through, or noncompliance incidents, if any, at the treatment plant which the Discharger suspects could be caused by commercial/industrial users of the system. The discussion shall also include a review of the applicable pollutant limitations to determine whether pre-treatment requirements may be necessary to prevent Upset, Interference, Pass Through, or noncompliance.
7. By **1 October 2001**, the Discharger shall submit a report prepared by a California Registered Civil Engineer or Engineering Geologist showing whether the facility is in compliance with Discharge Specifications B.10 and B.11, and if not, providing a workplan and a time schedule for attaining compliance and supporting hydraulic balance and capacity of each disposal pond.

CITY OF MARYSVILLE

MARYSVILLE WASTEWATER TREATMENT AND RECLAMATION FACILITY

YUBA COUNTY

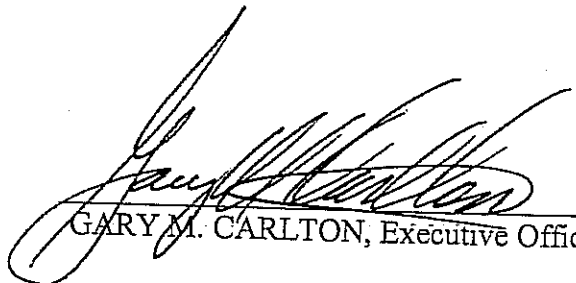
8. By 3 July 2002, the Discharger shall submit a technical report providing an assessment of alternatives, cost analysis, implementation plan, and proposed time schedule for attaining protection from the 100-year return frequency storm event. The report shall describe the capacity and expected life with supporting hydraulic balance for each disposal pond and the final disposal options for the waste solids in the ponds. The report shall be prepared by California registered civil engineer experienced in the field of wastewater treatment.
9. In accordance with Discharge Specification B.9, the Discharger shall implement an approved alternative project which will achieve compliance by 1 April 2006.
10. The Discharger shall comply with the Monitoring and Reporting Program (MRP) No. 5-01-071, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
11. The Discharger shall comply with the *STANDARD PROVISIONS AND REPORTING REQUIREMENTS FOR WASTE DISCHARGE REQUIREMENTS*, dated 1 March 1991, which are attached hereto and by reference are part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
12. The Discharger shall report promptly any material change or proposed change in the discharge character, location, or volume and/or the treatment train and/or the wastewater treatment facility components.
13. In the event of any change in control or ownership of land or waste discharge facilities described herein, The Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
14. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, The Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
15. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
16. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

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17. The Discharger shall immediately notify the Board by telephone whenever a violation or an adverse condition occurs as a result of the wastewater treatment operations or the disposal; a written confirmation providing a complete description of the event shall follow within two (2) weeks.
18. Operating personnel shall keep a copy of this Order at the City of Marysville wastewater treatment plant site for reference. Key operating personnel shall be familiar with its contents.
19. If reclaimed water is used for construction purposes, it shall comply with the most current edition of *Guidelines for Use of Reclaimed Water for Construction Purposes*. Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CFR, Division 4.
20. The Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 16 March 2001.


GARY M. CARLTON, Executive Officer

16 March 2001
DJL

Marysville.wdr.3

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM NO. 5-01-071

FOR
CITY OF MARYSVILLE
WASTEWATER TREATMENT AND RECLAMATION FACILITY
YUBA COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring domestic wastewater, treated effluent, wastewater storage ponds, land application areas, and groundwater. This MRP is issued pursuant to Water Code Section 13267. The Discharger shall not implement any changes to this MRP unless and until a revised MRP is issued by the Executive Officer. Regional Board staff shall approve specific sample station locations prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form. Field test instruments (such as those used to measure pH and dissolved oxygen) may be used provided that:

1. The operator is trained in proper use and maintenance of the instruments;
2. The instruments are field calibrated prior to each monitoring event;
3. The instruments are serviced and/or calibrated by the manufacturer at the recommended frequency; and
4. Field calibration reports are submitted as described in the "Reporting" section of the MRP.

INFLUENT MONITORING

Influent flow monitoring shall be performed at the headworks. Influent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Daily Flow	gpd	Meter	Continuously ¹	Monthly
Monthly Average Daily Flow	gpd	Calculated	Monthly	Monthly
BOD ²	mg/L	24-hr Composite	Monthly	Monthly

¹ Continuous monitoring requires daily meter reading or automated data acquisition.

² 5-day Biochemical Oxygen Demand.

EFFLUENT MONITORING
(Discharge to Wastewater Storage Ponds)

Effluent samples shall be collected before discharge to the wastewater storage ponds and shall be representative of the volume and nature of the discharge. Effluent monitoring shall include the following:

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<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
BOD	mg/L	Grab	Weekly	Monthly
Total Settleable Solids	mL/L	Grab	Weekly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
pH	Standard	Grab	Monthly	Monthly
Standard Minerals ¹	mg/L	Grab	Annually	Annually

¹ Standard Minerals shall include, at a minimum, the following elements/compounds: Boron, Calcium, Chloride, Magnesium, Potassium, Sodium, Sulfate, Total Alkalinity (including alkalinity series), and Hardness.

STORAGE POND MONITORING

Each of the storage ponds shall be sampled for the parameters specified below:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Freeboard	0.1 Feet	Measurement	Weekly	Weekly
Levee Condition	--	Observation	Weekly	Monthly
Odors	--	Observation	Weekly	Monthly
Dissolved Oxygen ¹	mg/L	Grab	Monthly	Monthly
pH	pH units	Grab	Monthly	Monthly

¹ Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet. Samples shall be collected between 0700 and 0900 hours.

TERTIARY RECYCLED WATER MONITORING (Wastewater Discharged to Soccer Fields)

Effluent discharged from the TTU shall be monitored as described below. In addition, monitoring of the land application areas shall be conducted **daily** (during operation) and the results shall be included in the monthly monitoring report. Evidence of erosion, land application area saturation, runoff, or the presence of nuisance conditions shall be noted in the report.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gallons	Continuous	Daily	Monthly
Total Coliform Organisms	MPN/100 mL	Grab	Daily	Monthly
Turbidity ¹	NTU	Continuous	Daily	Monthly
Acreage Applied ²	Acres	Calculated	Daily	Monthly
Application Rate ³	gal/acre-day	Calculated	Daily	Monthly
BOD	mg/L	Grab	Monthly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly

Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
BOD Loading Rate ³	lbs/acre•day	Calculated	Monthly	Monthly
Total Nitrogen Loading Rate ³	lbs/month	Calculated	Monthly	Monthly
TDS Loading Rate ³	lbs/month	Calculated	Monthly	Monthly

¹ The Discharger shall report the daily average turbidity as well as the total amount of time each day that the turbidity exceeded 5 NTU and the total amount of time each day that the turbidity exceeded 10 NTU.

² Land application areas shall be identified.

³ For each land application area.

RECYCLED WATER MONITORING (Wastewater Discharged to Walnut Orchard)

Effluent discharged to the walnut orchard shall be monitored as described below. In addition, monitoring of the land application areas shall be conducted **daily** (during operation) and the results shall be included in the monthly monitoring report. Evidence of erosion, land application area saturation, runoff, or the presence of nuisance conditions shall be noted in the report.

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>	<u>Reporting Frequency</u>
Flow	Gallons	Continuous	Daily	Monthly
Total Coliform Organisms	MPN/100 mL	Grab	Daily	Monthly
Acreage Applied ¹	Acres	Calculated	Daily	Monthly
Application Rate ²	gal/acre•day	Calculated	Daily	Monthly
BOD	mg/L	Grab	Monthly	Monthly
Total Dissolved Solids	mg/L	Grab	Monthly	Monthly
Nitrate as Nitrogen	mg/L	Grab	Monthly	Monthly
Total Kjeldahl Nitrogen	mg/L	Grab	Monthly	Monthly
BOD Loading Rate ²	lbs/acre•day	Calculated	Monthly	Monthly
Total Nitrogen Loading Rate ²	lbs/month	Calculated	Monthly	Monthly
TDS Loading Rate ²	lbs/month	Calculated	Monthly	Monthly

¹ Land application areas shall be identified.

² For each land application area or sub-area.

GROUNDWATER MONITORING

Prior to construction and/or sampling of any groundwater monitoring wells, the Discharger shall submit plans and specifications to the Board for review and approval. Once installed, all new wells shall be added to the MRP and shall be sampled and analyzed according to the schedule below.

Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged at least three well volumes until temperature, pH and electrical conductivity have stabilized. Depth to groundwater shall be measured, and groundwater elevation calculated to the nearest 0.01 feet. Samples

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shall be collected using standard EPA methods. Groundwater monitoring shall include, at a minimum, the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling and Reporting Frequency</u>
Depth to Groundwater	0.01 Feet	Measurement	Quarterly
Groundwater Elevation	0.01 Feet	Measurement	Quarterly
Gradient	Ft./Ft.	Calculated	Quarterly
Gradient Direction	Degrees	Calculated	Quarterly
Total Dissolved Solids	mg/L	Grab	Quarterly
Nitrate as Nitrogen	mg/L	Grab	Quarterly
Total Kjeldahl Nitrogen	mg/L	Grab	Quarterly
PH	pH units	Grab	Quarterly
Total Coliform Organisms	MPN/100 ml	Grab	Quarterly
Trihalomethanes	µg/L	Grab	Quarterly
Boron	mg/L	Grab	Quarterly
Iron	mg/L	Grab	Quarterly
Manganese	mg/L	Grab	Quarterly
Chloride	mg/L	Grab	Quarterly
Sodium	mg/L	Grab	Quarterly
Standard Minerals ¹	mg/L	Grab	Annually

¹ Standard Minerals shall include, at a minimum, the following elements/compounds: Calcium, Magnesium, Potassium, Sulfate, Total Alkalinity (including alkalinity series), and Hardness.

SLUDGE MONITORING

Prior to the removal of sludge from any pond, a composite sample shall be collected in accordance with EPA's POTW Sludge Sampling and Analysis Guidance Document (August 1989) and tested for the following metals: cadmium, copper, nickel, chromium, lead, and zinc.

Sampling records shall be retained for a minimum of five years. A log shall be kept of sludge quantities generated and of handling and disposal activities. The frequency of entries is discretionary; however, the log should be complete enough to serve as a basis for part of the annual report.

REPORTING

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Regional Board.

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As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

A. Monthly Monitoring Reports

Daily, weekly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, storage pond, tertiary recycled water to soccer fields, and recycled water to orchard monitoring;
2. A comparison of monitoring data to the discharge specifications and an explanation of any violation of those requirements. Data shall be presented in tabular format; and
3. If requested by staff, copies of laboratory analytical report(s);
4. A calibration log verifying calibration of all hand-held monitoring instruments and devices used to comply with the prescribed monitoring program.

B. Quarterly Monitoring Reports

The Discharger shall establish a quarterly sampling schedule for groundwater monitoring such that samples are obtained approximately every three months. Quarterly monitoring reports shall be submitted to the Board by the **1st day of the second month after the quarter** (i.e. the January-March quarterly report is due by May 1st) and may be combined with the monthly report. The Quarterly Report shall include the following:

1. Results of groundwater monitoring;
2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for the groundwater monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. The narrative shall be supported by field logs for each well documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged;
3. Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any;
4. A narrative discussion of the analytical results for all groundwater locations monitored including spatial and temporal trends, with reference to summary data tables, graphs, and appended analytical reports (as applicable);
5. A comparison of monitoring data to the groundwater limitations and an explanation of any violation of those requirements;
6. Summary data tables of historical and current water table elevations and analytical results;

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7. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and any other sampling stations, and groundwater elevation contours referenced to mean sea level datum;
8. Copies of laboratory analytical report(s) for groundwater monitoring.

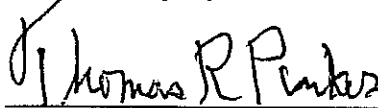
C. Annual Report

An Annual Report shall be prepared as the fourth quarter monitoring report. The Annual Report will include all monitoring data required in the monthly/quarterly schedule. The Annual Report shall be submitted to the Regional Board by 1 **February** each year. In addition to the data normally presented, the Annual Report shall include the following:

1. The contents of the regular groundwater monitoring report for the last sampling event of the year;
2. If requested by staff, tabular and graphical summaries of all data collected during the year;
3. An evaluation of the groundwater quality beneath the wastewater treatment facility;
4. A discussion of compliance and the corrective actions taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements;
5. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program;
6. Summary of information on the disposal of sludge and/or solid waste;
7. The results from annual monitoring of the effluent and groundwater;
8. The results from any sludge monitoring required by the disposal facility;
9. A copy of the certification for each certified wastewater treatment plant operator working at the facility and a statement about whether the Discharger is in compliance with Title 23, CCR, Division 3, Chapter 26; and
10. A forecast of influent flows predicted for the next year.

The Discharger shall begin the implementation of the above monitoring program on 1 November 2004.

Ordered by:



THOMAS R. PINKOS, Executive Officer

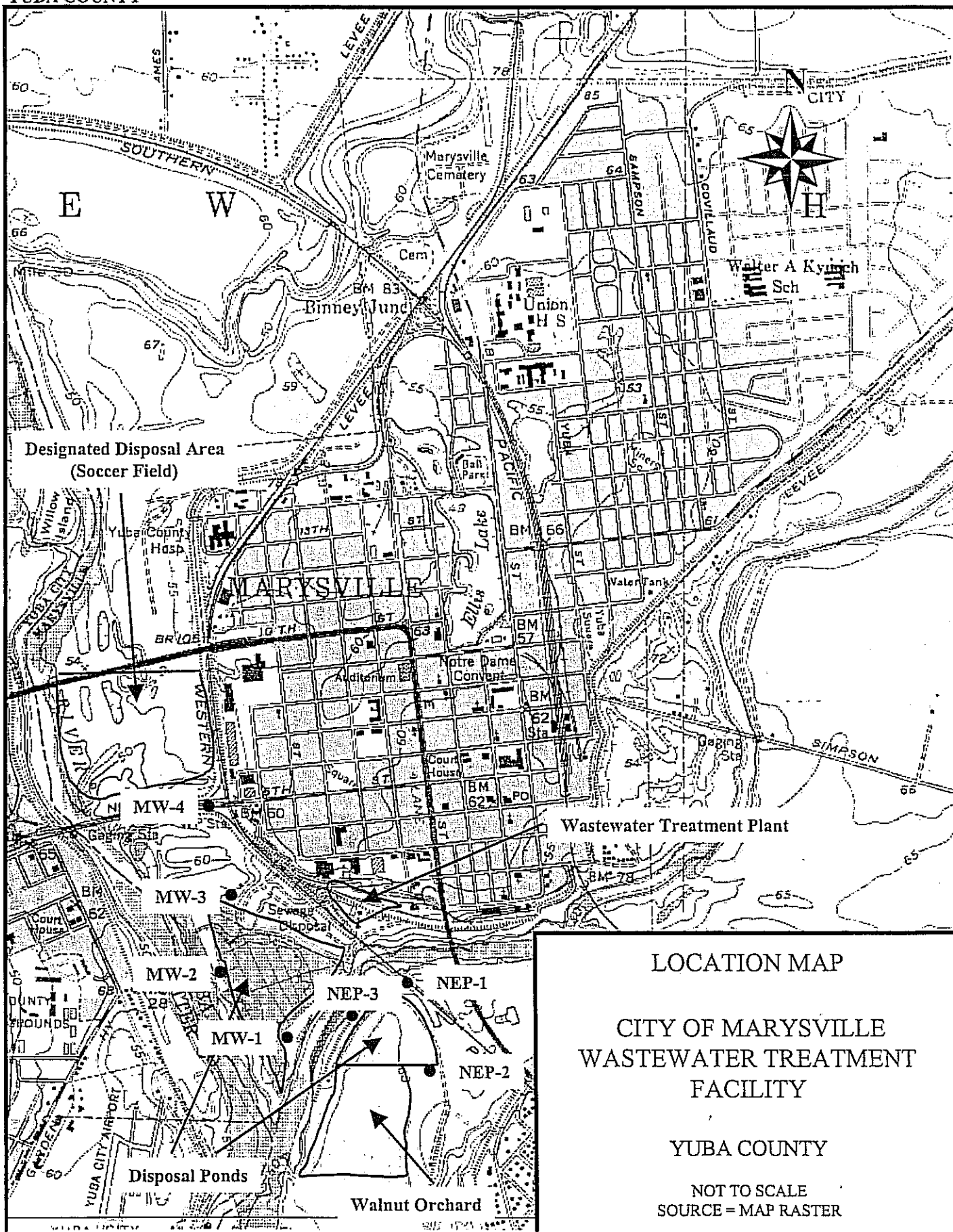
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INFORMATION SHEET
CITY OF MARYSVILLE
WASTEWATER TREATMENT AND RECLAMATION FACILITY
YUBA COUNTY

The City of Marysville Wastewater Treatment and Reclamation Facility is owned and operated by the City in Yuba County. The system services the City of Marysville and surrounding area. Previous Waste Discharge Requirements (WDRs) were set forth in Order Nos. 89-210 and 95-093. The existing WDRs Order No. 95-093 is being updated to reflect current State regulations, Board policies, and the addition of an off-site recycled water project which will use treated and disinfected wastewater for walnut orchard irrigation.

The Dischargers facilities consist of preliminary treatment, primary clarification, trickling filters, secondary clarification, anaerobic digestion, and sludge drying beds. The design capacity of the secondary plant and pond system is 1.7 MGD. Disposal of effluent is accomplished via discharge of approximately 1.6 million gallons per day (MGD) to evaporation/percolation ponds and the designated water reclamation area, which is the walnut orchard. The new tertiary treatment unit (TTU) consists of clarification, filtration, chlorination, and holding ponds. The design capacity of the TTU is 0.8 MGD. The treated effluent from the TTU will be Title 22 quality water with a weekly median of 2.2 MPN/110ml total coliform organisms for proposed spray irrigation disposal on the soccer fields.

Effluent limits prescribed in the waste discharge requirements were designed to protect the public from odor nuisance conditions and minimize the impact in the event that runoff from the application areas enters surface waters. Regarding the discharge of secondary effluent to the evaporation/percolation ponds, the maximum allowable 30-day average and daily, five-day 20°C Biological Oxygen Demand was set at 40 and 80 mg/l, respectively. The maximum allowable 30-day average and daily Settleable Solids was set at 0.2 and 0.5 ml/l, respectively. Regarding the discharge of tertiary effluent to the designated reclamation disposal areas, the maximum allowable 30-day average and daily, five day 20°C Biological Oxygen Demand was set at 10 and 30 mg/l, respectively. The maximum allowable weekly median and daily maximum average Total Coliform Organisms was set at 2.2 and 23 MPN/100 ml, respectively.

The disposal ponds are protected from a 10-year flood event. The Order requires the Discharger to submit a technical report providing an evaluation of alternatives, a cost analysis, an implementation plan, and a proposed time schedule for attaining compliance for the 100-year return frequency storm event. The plan would include protecting the designated wastewater disposal areas, except for the walnut orchard and soccer fields, from inundation or washout from flood and 100-year return frequency storm over the entire winter season. Monitoring wells are installed in the vicinity of the disposal ponds to monitor the effects of the ponds on the groundwater. The proposed WDRs require quarterly sampling of those wells.

The beneficial uses of the Feather and Yuba Rivers are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; and preservation and enhancement of fish, wildlife and other aquatic resources. Depth to ground water at the site is about 18 - 34 feet. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply. Surface water drainage from the facility is to the Feather River and the Yuba River.

